



Methylation-based aging in a salmonid

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Introduction

- Tony Kess
- Bioinformatics Biologist
Fisheries and Oceans
Canada – DFO Salmonids
- GenARCC scientific
computing lead
- tony.kess@dfo-mpo.gc.ca



Acknowledgements in advance

Acknowledgements:

Ian Bradbury

Bradbury lab

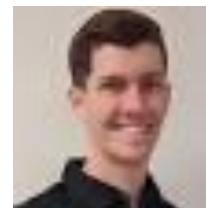
DFO NL, ARCTIC, & MAR
teams

GRDI GenARCC teams

Bentzen lab

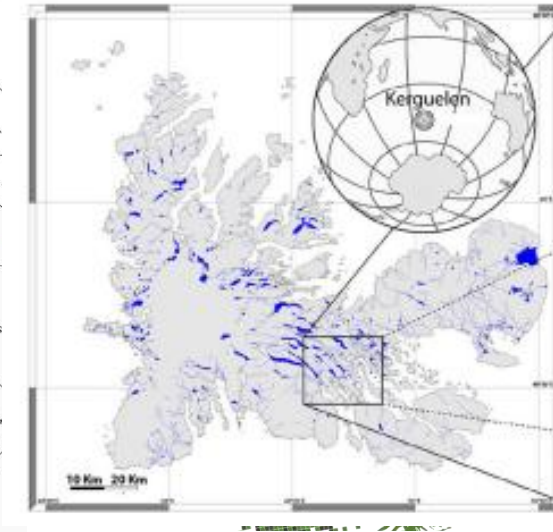
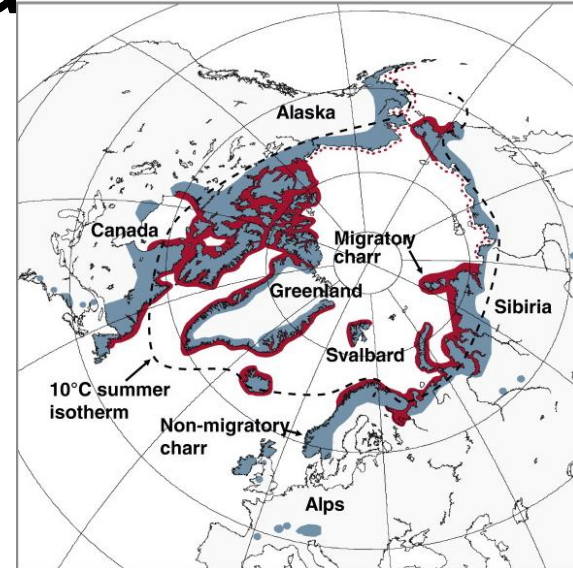
Fleming lab

Ruzzante lab



Background – Arctic Charr

- The northernmost freshwater fish!
- And the southernmost?
- Perhaps the most variable vertebrate, high habitat diversity
- Supports food fisheries for northern communities



Piscivorous morph



Deep-water morph



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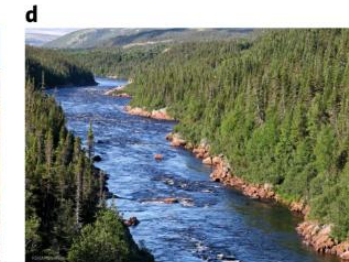
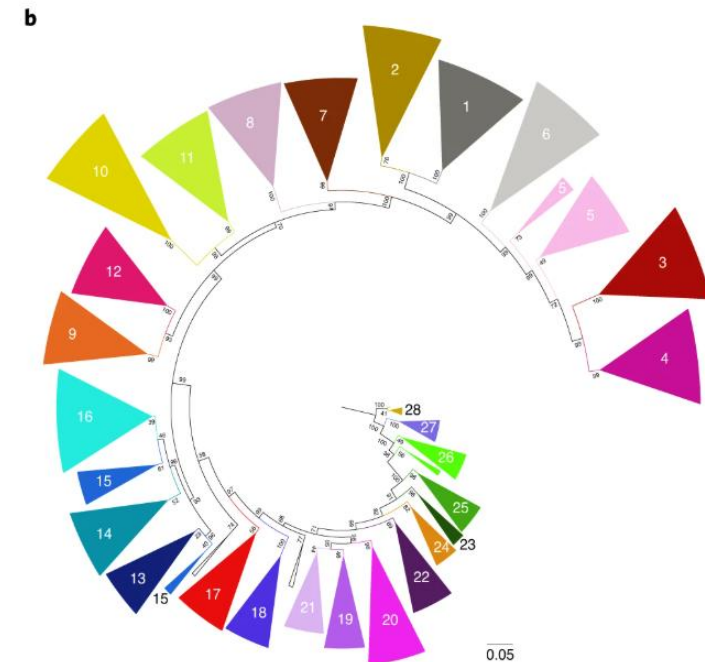
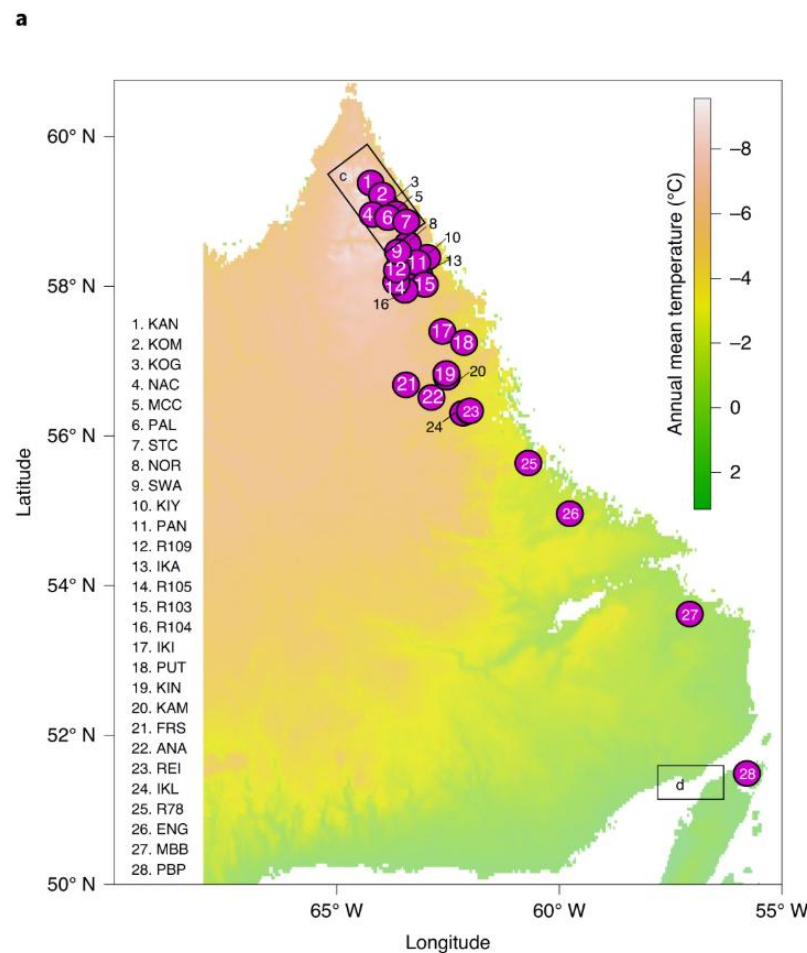
The most variable vertebrate on Earth

Published: 28 December 2013

Volume 53, pages 781–791, (2013) [Cite this article](#)

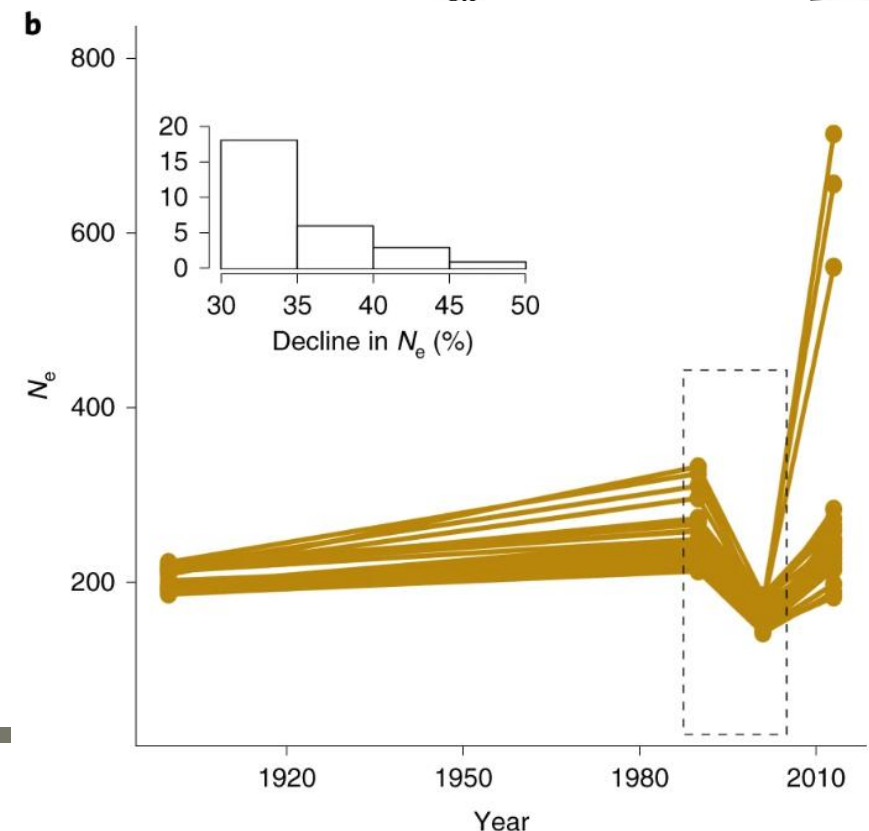
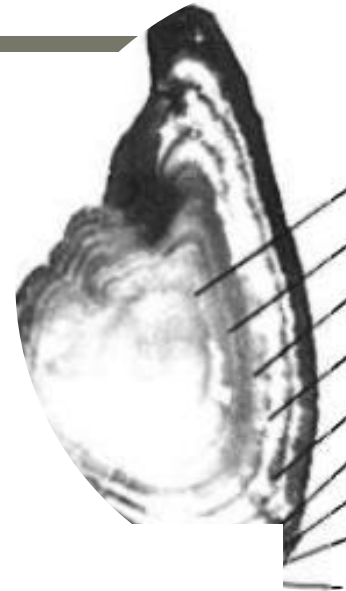
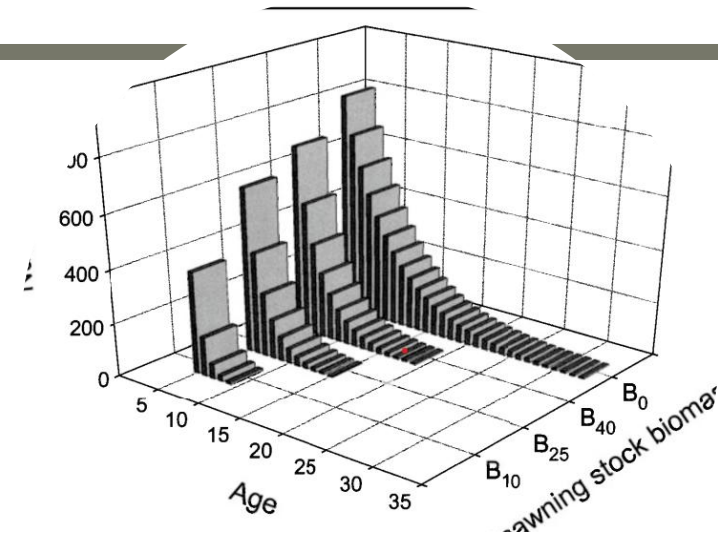
Arctic Charr – stock information

- Highly structured populations with local genetic variation
- Climate change risk with habitat loss in southern range
- **Minimal demographic information on most populations**



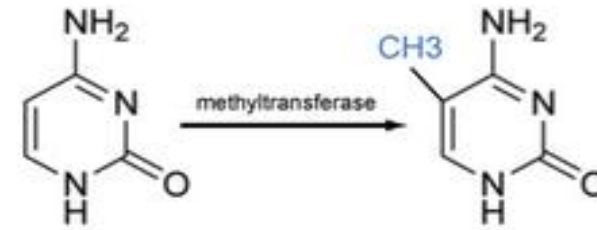
Stock assessment challenges

- Understanding age structure important for stock assessment, predicting future fishery productivity
- Currently measured with otolith-based aging
- Labour and skill intensive, **lethal**
- Small N_e populations – important to minimize lethal sampling where possible

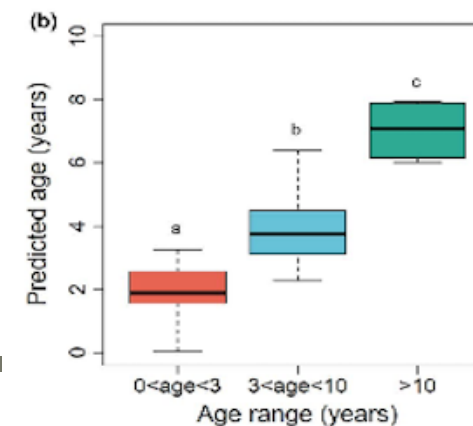
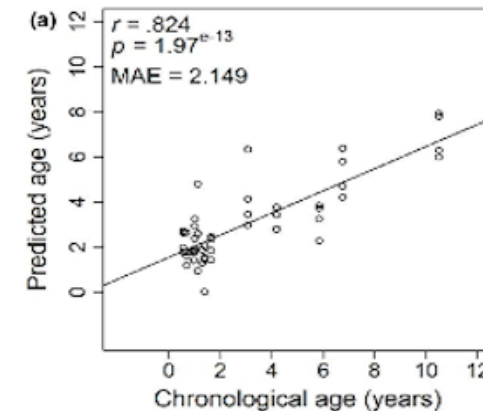


Methylation-based aging

- Changes to DNA that increase with age have been used to predict chronological age
- High throughput, high accuracy, non-lethal aging
- Used in other fish species for age estimation with high predictive power, sample sizes ~100-200 individuals



DNA methylation



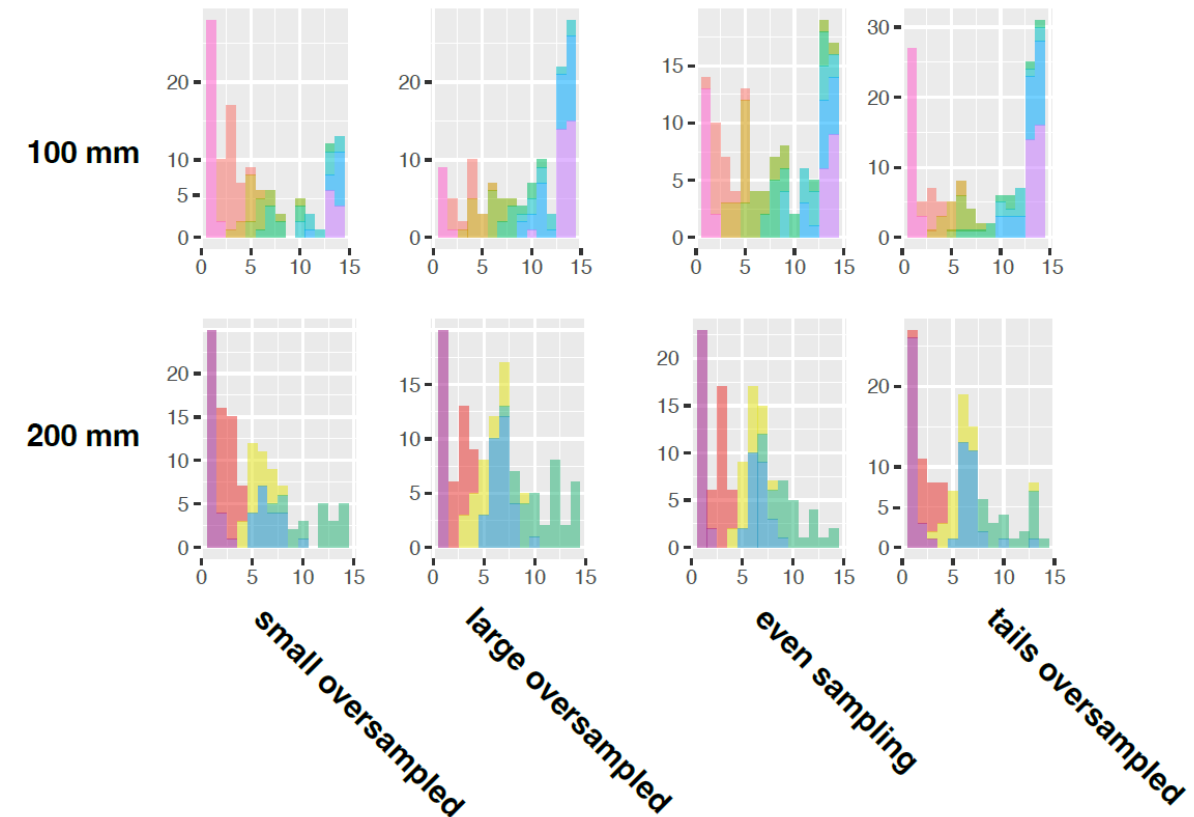
Project background

- 3 Year DFO CSRF project funded 2022-2025
- Develop methylation markers for aging Arctic Charr
- Focused on 2 sites in 2 regions (Labrador & Arctic)
- Collection goal ~200 fish per site



Field work and collection strategy

- Lethal sampling and otolith extraction
- Ages unknown, used Nain length x age data to project expected ages with size sampling scheme



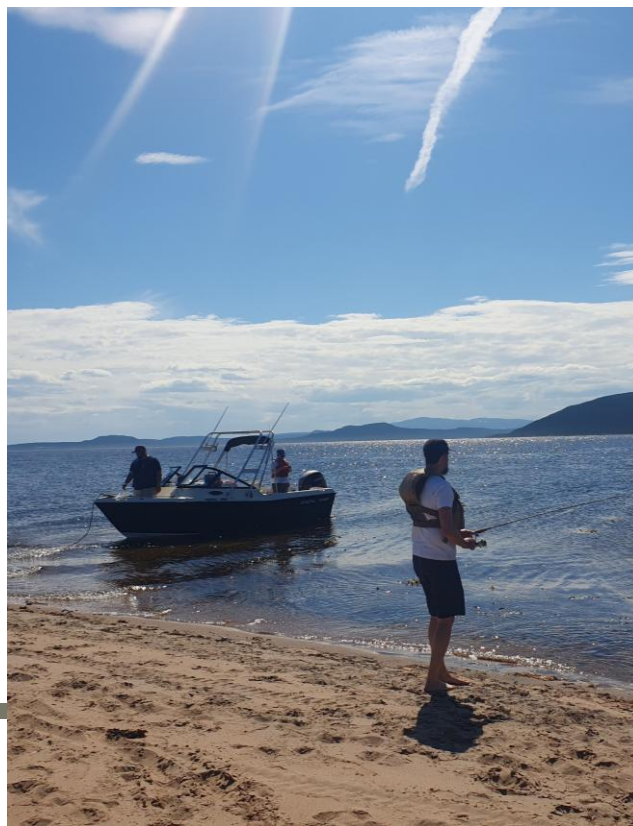
2023 Field update

Year 2: Field sampling in Muddy Bay/Cartwright to complete sample composition for aging study, July 9 - 17





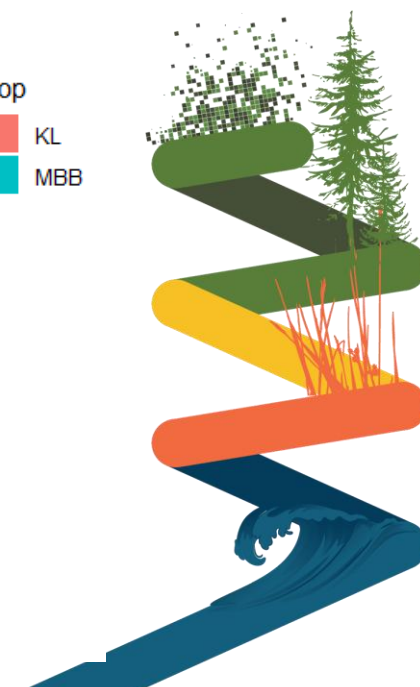
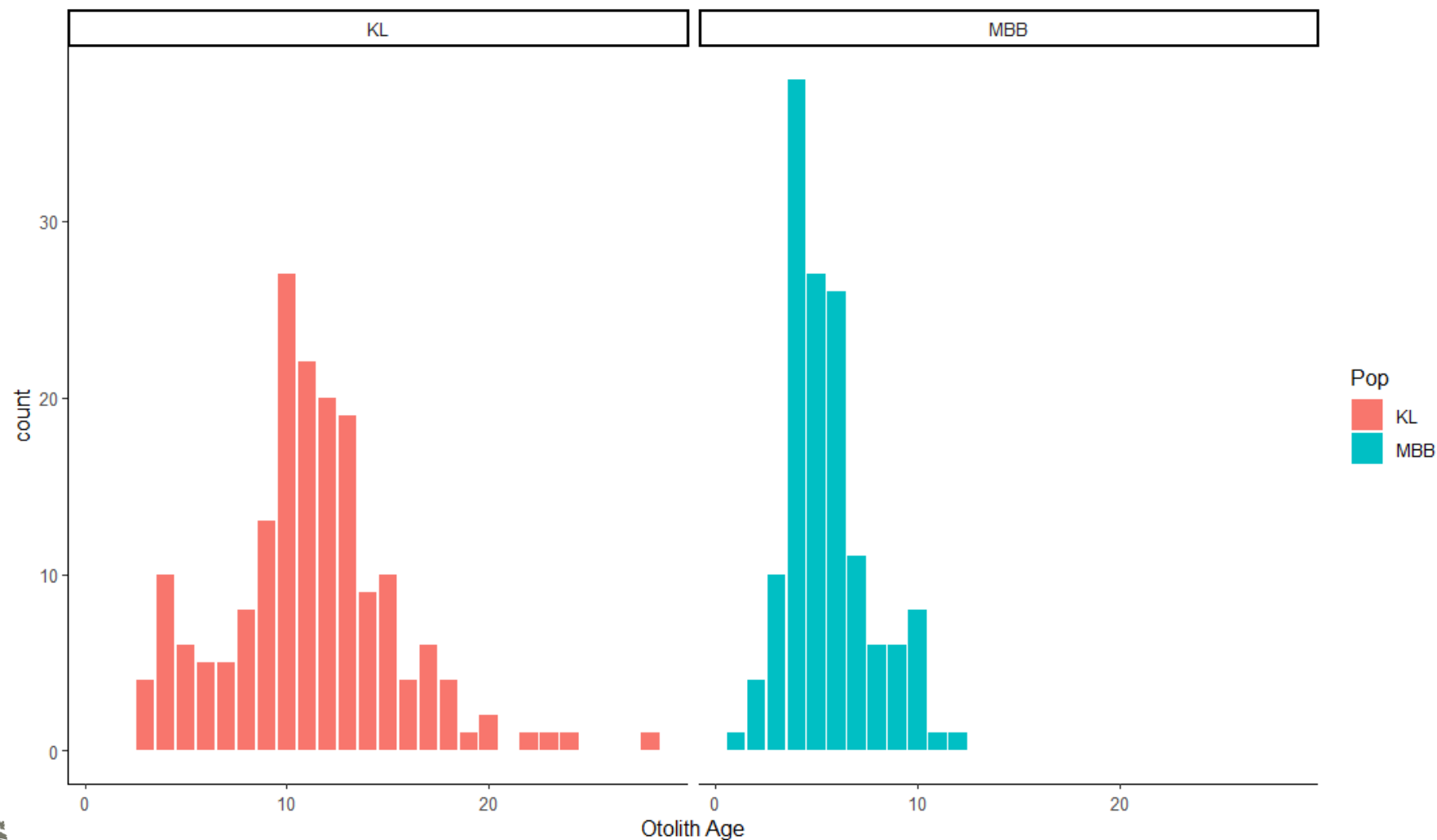
2023 Field update



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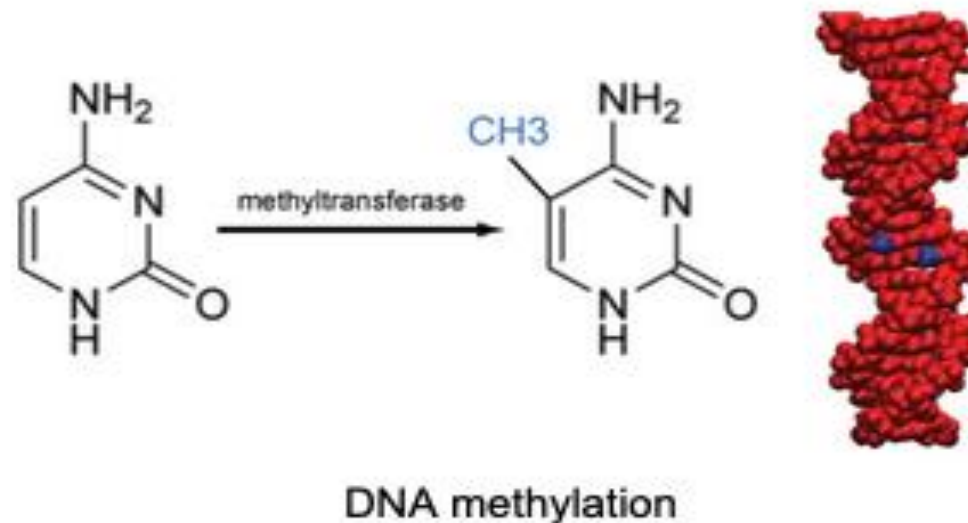


Age distribution



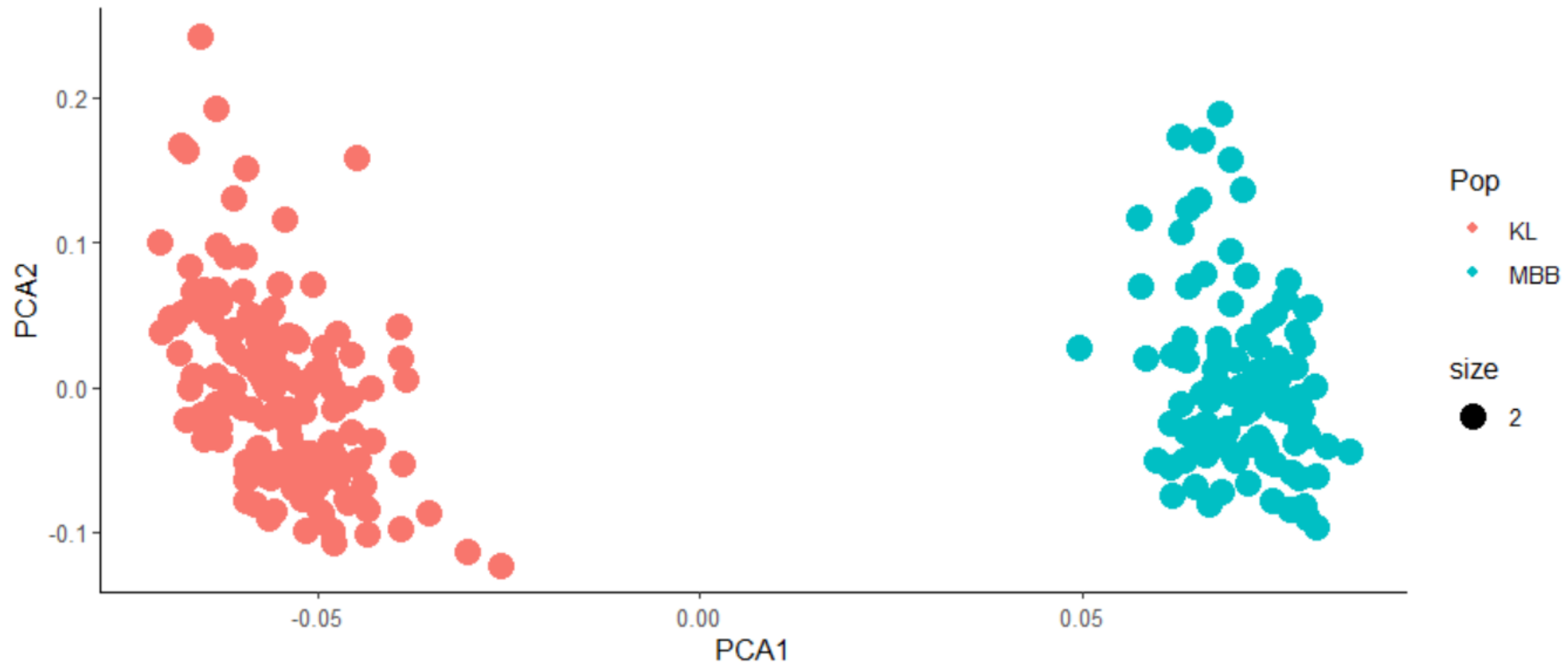
Sequencing

- Reduced representation bisulfite sequencing
- Incomplete genomic coverage
- Enriched towards methylation sites
- 8000 methylation markers across all individuals
- 184 KL
- 141 MBB



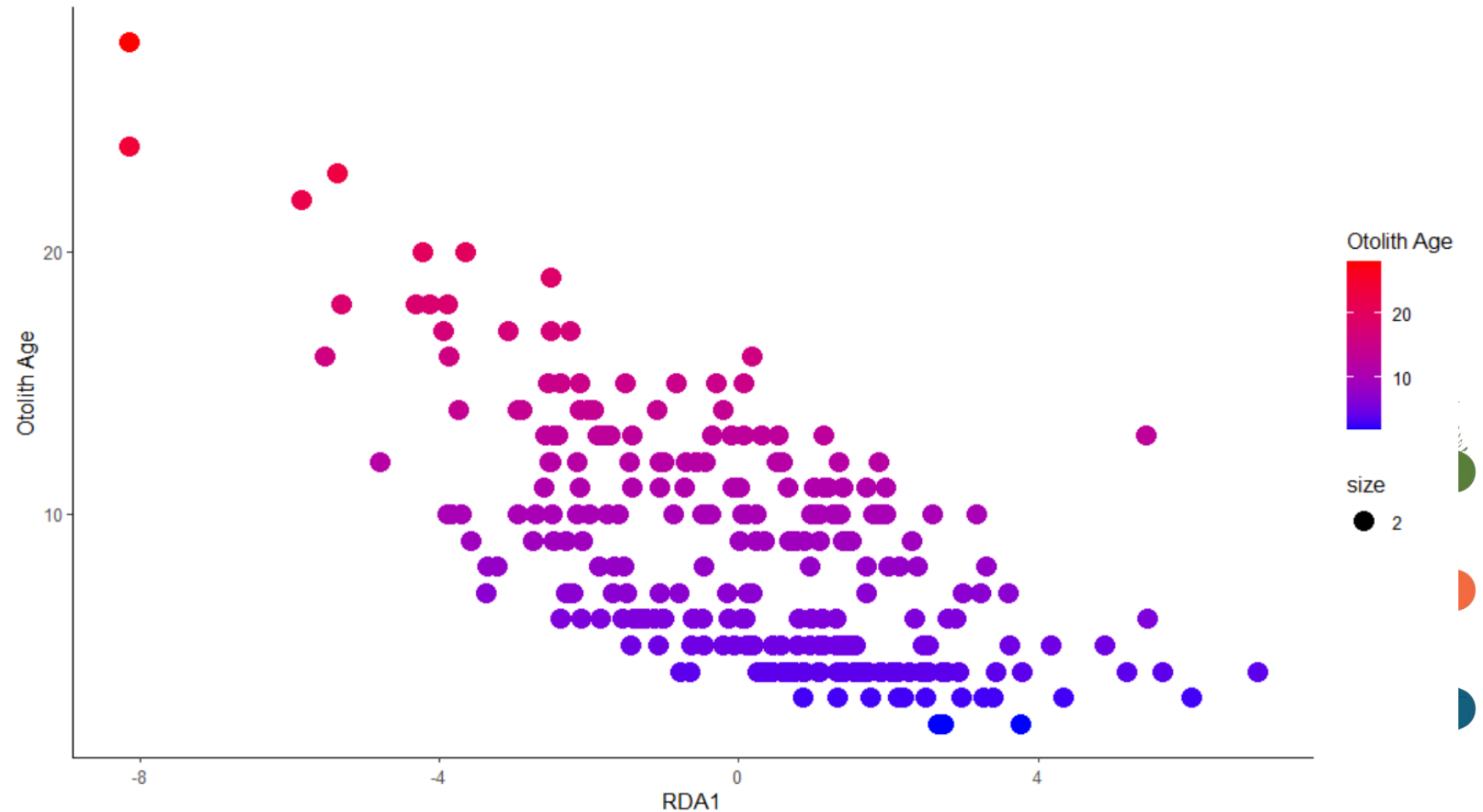
Methylation - PCA

- Main variation in methylation data is location



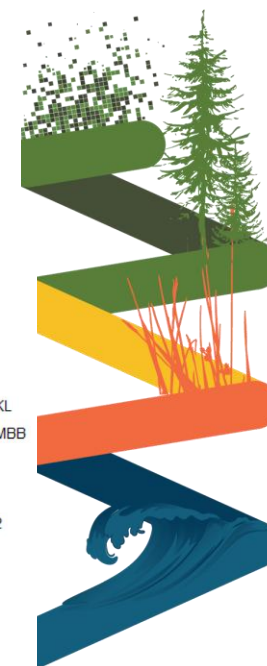
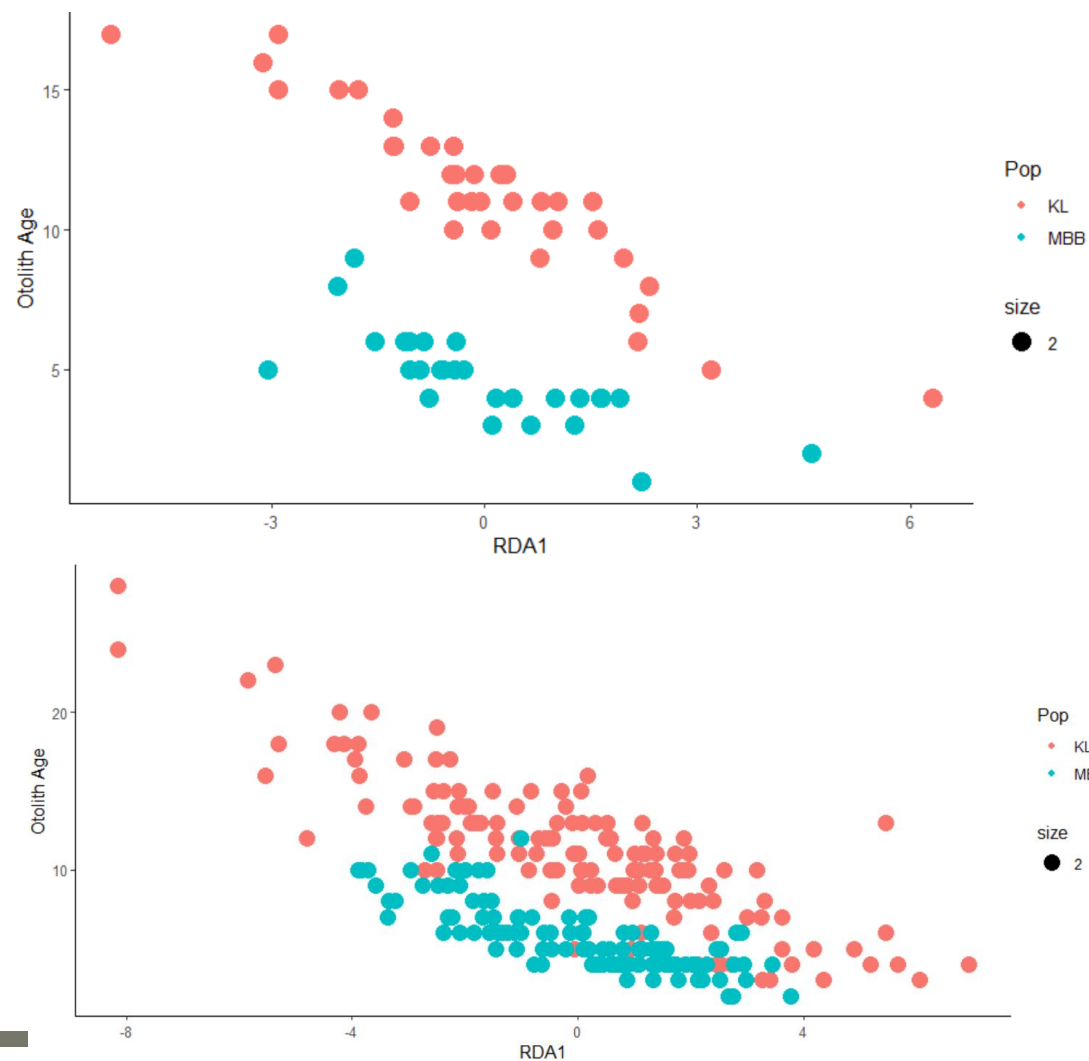
Ordination of methylation by age

- Training set – 80% of individuals
- Redundancy analysis – constrained ordination to project age variation by methylation variation
- $R^2 = 0.4225$



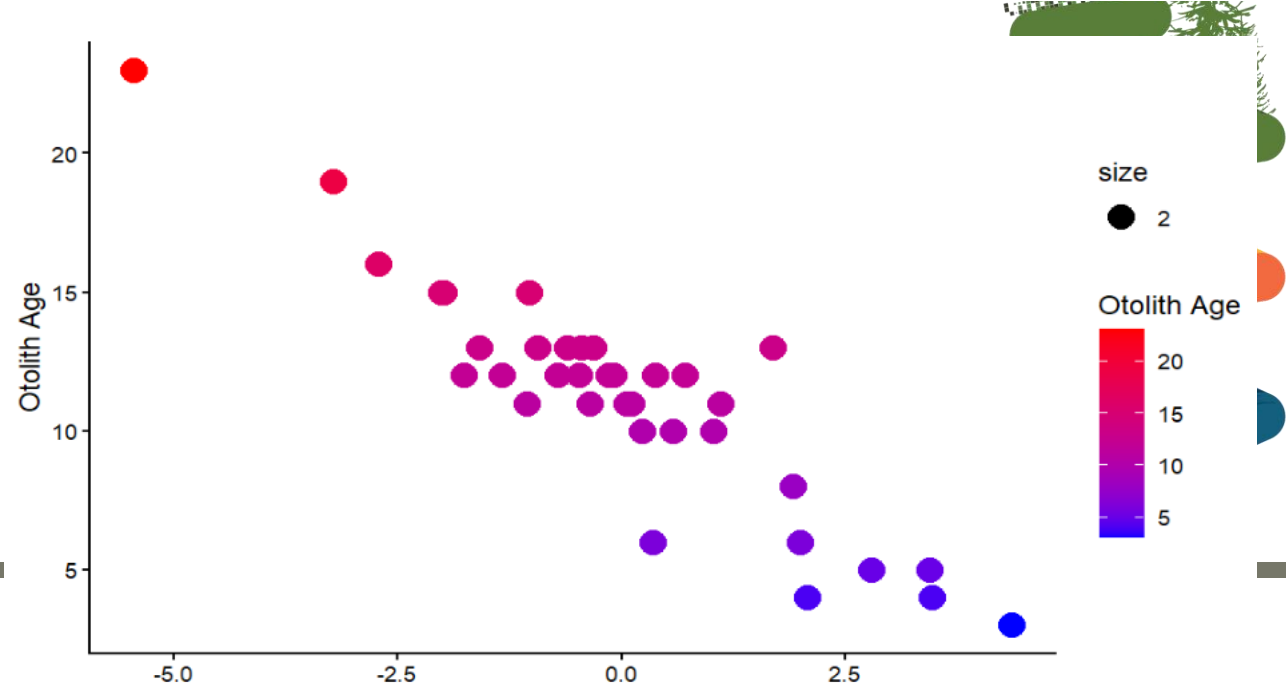
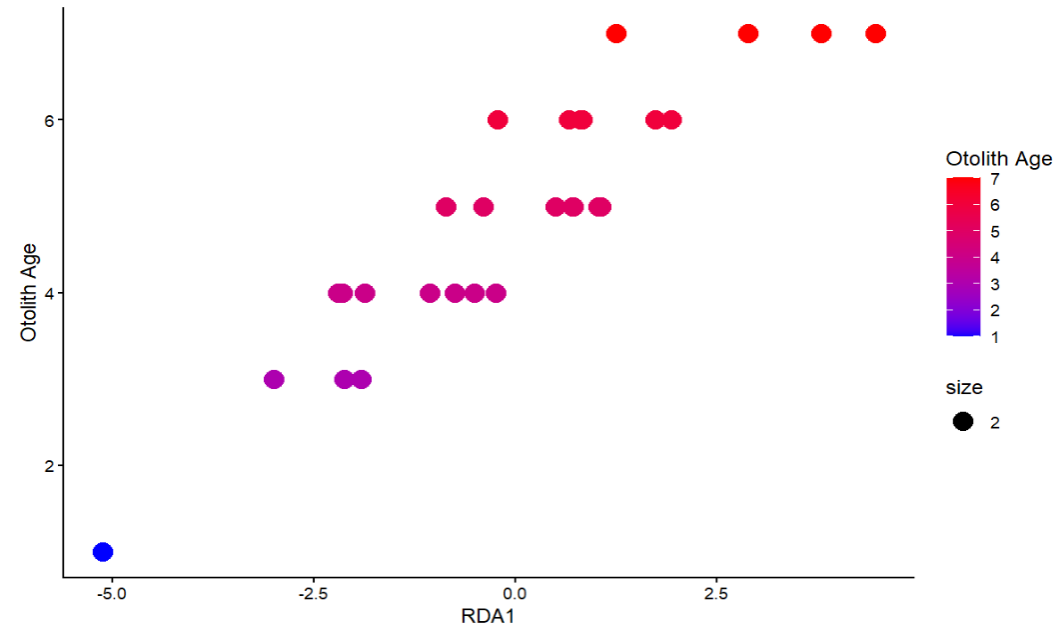
Ordination of methylation by age

- Test set – 20% of held out individuals
- Used methylation outliers, highest RDA score
- Not so good
- Population stratification effects!



Accounting for population structure

- Test set – 20% of held out individuals
- Population specific models regain performance
- $R^2 > 0.8$



Conclusions and next steps

- Methylation can be used to predict age in Arctic Charr!
- Genetic structuring in salmonids indicates need for more sampling, or population specific models
- Now expanding to more rivers across Labrador
- Interest in methylation beyond an age-structure estimation tool, utility may not be just in accuracy of age estimate but multimodal data per sample, e.g. thermal stress, disease

